REAL TIME SIGN LANGUAGE DETECTION

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Abstract: This paper presents the development of a real-time sign language recognition system using computer vision, machine learning, and deep learning techniques. The primary objective is to reduce communication barriers for the hearing and speech-impaired community by accurately recognizing sign language gestures from video sequences. The proposed framework utilizes a diverse dataset of sign language gestures captured under varying conditions and by multiple signers. Advanced preprocessing techniques are applied to extract meaningful features from video frames, including hand movements, facial expressions, and body posture. To effectively capture temporal dependencies in sequential data, a Long Short-Term Memory (LSTM) based recurrent neural network architecture is employed. The model is trained using labeled gesture sequences, while transfer learning and data augmentation techniques are incorporated to improve robustness and generalization. Experimental results demonstrate that the proposed system can accurately recognize a wide range of sign language gestures in real time, highlighting its potential as an assistive communication tool. ***Keywords:*** Sign Language Recognition, Computer Vision, LSTM, Transfer Learning, Data Augmentation, Real-Time Detection

1. *Introduction*

Real-time sign language recognition has become a significant research area aimed at reducing the communication barrier between hearing- and speech-impaired individuals and the general population. Sign language is the primary mode of communication for millions of people worldwide; however, the limited understanding of sign language among the broader community often creates challenges in daily communication, education, employment, and access to public services. With the rapid advancement of computer vision, machine learning, and deep learning technologies, new opportunities have emerged to develop intelligent systems capable of automatically interpreting sign language gestures with high accuracy and speed.

The proposed real-time sign language detection system seeks to develop a robust and efficient framework that can observe, analyze, and recognize sign language gestures using advanced computer vision techniques and neural network models. The system continuously tracks hand movements, extracts meaningful spatial and temporal features, and applies deep learning algorithms to classify gestures in real time. By translating recognized gestures into text or speech, the system enables seamless and instant interaction between sign language users and the broader community. Such assistive technology has the potential to greatly enhance accessibility and inclusivity in education, healthcare, workplace environments, and everyday social communication, ultimately improving the quality of life for individuals with hearing and speech impairments.

2. Related Work

Sign language recognition has gained significant attention in recent years, with numerous studies exploring the use of deep learning and computer vision techniques. Mali et al. (2023) developed a sign language recognition system using MediaPipe Holistic and Long Short-Term Memory (LSTM) networks. Their approach captured pose, face, and hand keypoints with low latency and achieved an accuracy of 98.5% for commonly used American Sign Language words. Similarly, Mhatre et al. (2022) proposed a real-time sign language detection system based on LSTM that recognized commonly used Marathi sign gestures and generated both text and speech output, achieving training accuracy between 90% and 96%.

Aparna et al. (2020) introduced a deep learning approach combining Convolutional Neural Networks (CNN) and LSTM for Indian Sign Language recognition. CNN was used for feature extraction while LSTM captured temporal dependencies, resulting in improved accuracy. Sundar et al. (2022) further explored MediaPipe-based hand landmark detection with LSTM for American Sign Language alphabet recognition and reported an accuracy of 99%.

Deshpande et al. (2022) proposed a real-time Indian Sign Language recognition system using CNN architectures to translate gestures into words and speech. Their model achieved 98% accuracy on a dataset containing digits, letters, and common words. Pathak et al. (2022) used transfer learning with SSD MobileNet V2 for gesture classification and achieved an accuracy of 70–80% under real-world conditions.

Kothadiya et al. (2022) developed DeepSign, a hybrid deep learning model using LSTM and GRU networks for Indian Sign Language recognition, achieving approximately 97% accuracy. Kodandaram et al. (2021) highlighted the importance of CNN-based deep learning approaches for recognizing both static and dynamic gestures and converting them into speech output.

Sharma and Singh (2021) created a large Indian Sign Language dataset and developed a CNN-based recognition system that achieved up to 99.52% accuracy. Deep et al. (2022) proposed a real-time sign detection system combining OpenCV, MediaPipe, TensorFlow, and LSTM to enable live webcam-based gesture recognition. Although these studies demonstrate significant progress, challenges remain in achieving robust real-time performance across diverse environments and users. This motivates the development of the proposed LSTM-based real-time sign language detection system. Here is the rewritten and improved Methodology, System Implementation, LSTM Usage and Conclusion in proper research-paper style.

3. Methodology

The proposed system is designed to recognize Indian Sign Language (ISL) gestures in real time and convert them into meaningful text and speech responses. The methodology integrates computer vision, deep learning, and sequence modeling to build a complete end-to-end communication framework. The overall workflow begins with capturing live video input, followed by gesture detection and tracking, feature extraction, sequence modeling, translation, and response generation. Initially, video frames are captured through a webcam and processed using computer vision techniques to detect and track hand and body movements. From these frames, spatial and temporal features such as hand landmarks, motion trajectories, and gesture dynamics are extracted. These features are then organized into sequential data, which forms the input for the deep learning model. To effectively model the temporal nature of sign language, a Long Short-Term Memory (LSTM) network is employed. The LSTM learns patterns from the sequential gesture data and maps them to corresponding textual representations of Indian Sign Language. The recognized text is then translated into English and converted into speech to facilitate real-time communication.

Finally, to enable bidirectional interaction, the generated response can be converted back into sign language and displayed using a virtual avatar. This integrated pipeline ensures accurate gesture recognition, real-time processing, and seamless communication between sign language users and the general population.

4. System Implementation

The system implementation involves multiple stages that work together to enable real-time gesture recognition and bidirectional communication.

The first stage focuses on Indian Sign Language gesture detection, where hand and body movements are tracked from video frames using computer vision techniques. Relevant visual cues such as hand shape, joint positions, and motion trajectories are captured to identify gestures accurately. After gesture detection, feature extraction is performed to obtain meaningful spatial and temporal information from the tracked movements. These features represent the dynamic characteristics of gestures and form the input for the deep learning model. The extracted feature sequences are then processed using a Long Short-Term Memory (LSTM) network. The LSTM model learns temporal dependencies and patterns within gesture sequences and maps them to corresponding Indian Sign Language text representations. Once the ISL text is generated, translation techniques are applied to convert the recognized signs into English text. The translated text is then used to generate an appropriate response. To enable bidirectional communication, the generated English text is converted back into Indian Sign Language text using a reverse translation mechanism. To visually represent the response, corresponding gesture videos are retrieved from a predefined dataset. Features are extracted from these videos and used to animate a 3D avatar that performs the required sign language gestures, enabling an interactive and inclusive communication system.

5. Role of LSTM in Sign Language Detection

Long Short-Term Memory networks play a crucial role in capturing the temporal dependencies present in sign language gestures. Unlike traditional recurrent neural networks, LSTMs are designed to overcome the limitations of vanishing gradients and effectively model long-term dependencies. In this system, the extracted gesture features form a time-series sequence, making LSTM an ideal choice for processing the data. The memory cell in LSTM enables the model to retain contextual information from earlier frames, allowing it to understand the evolving dynamics of gestures over time. This capability is essential because sign language gestures vary in duration, complexity, and sequence. LSTM networks also adapt to variable-length gesture sequences, which is critical for modeling diverse sign language expressions. By learning temporal patterns and relationships, the model accurately maps gesture sequences to textual representations and improves recognition performance.

Conclusion

Recent advancements in computer vision and deep learning have significantly accelerated the development of real-time sign language recognition systems. The proposed framework demonstrates how modern AI techniques can effectively reduce the communication gap between

the hearing- and speech-impaired community and the general population. By integrating gesture detection, feature extraction, LSTM-based sequence modeling, translation, and avatar-based response generation, the system provides a comprehensive and inclusive communication solution capable of real-time interaction. Despite these promising results, several practical challenges remain, including variations in lighting conditions, complex backgrounds, signer diversity, and limited large-scale datasets. Future work will focus on improving model robustness through larger and more diverse datasets, enhancing real-time performance, and expanding the system to support multiple sign languages. These improvements will further strengthen the scalability and real-world applicability of intelligent sign language recognition systems.

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